

ADVANCED COMPUTER SCIENCE EXAMINATION

Course: CS401 Distributed Systems & Algorithms

Total Marks: 100 | Time: 3 Hours

SECTION A: MULTIPLE CHOICE QUESTIONS

1. Which sorting algorithm guarantees an $O(n \log n)$ worst-case time complexity?
 - (A) Quick Sort
 - (B) Merge Sort
 - (C) Bubble Sort
 - (D) Selection Sort

2. What is the primary purpose of the two-phase commit (2PC) protocol in distributed databases?
 - (A) Consensus on replication order
 - (B) Atomic transaction commitment across distributed nodes
 - (C) Deadlock prevention in shared locks
 - (D) Network partition detection

SECTION B: DESCRIPTIVE QUESTIONS

3. Explain Dijkstra algorithm for finding single-source shortest paths in a directed weighted graph.
Provide the pseudocode and analyze its asymptotic complexity using a binary min-heap.
4. Differentiate between optimistic and pessimistic concurrency control mechanisms in database management systems.